

latest and more path-breaking developments in this field was that of the brain-controlled wheelchair designed to help paralyzed people gain mobility by using nothing but their thoughts to control the wheelchair's movement.

Other examples of Brain-Computer Interfaces include the Epoc headset designed by Emotiv. Featured in numerous tech magazines and even on a TEDx talk, the Epoc headset is a wearable that can detect EEG signals in 14 high-resolution channels and translate this raw data to commands that can be passed to a software for applicative uses.

The potential for such technology when it comes to wearables is self-explanatory. It is especially useful when you think of potential to connect multiple smart devices to such an interface, allowing you to control items like appliances and electronics to possibly even more complicated applications such as driving or translating thoughts to text. Envision a future where you can operate a series of interconnected/interdependent devices using thoughts alone.

Tangible User Interface (TUI)

We've already talked about the interaction of real-world objects with virtually designed ones in the discussion of AR-based interfaces. However, the digital objects in AR are not tangible and will always remain virtual. We are but grasping at air, and it is a well-known fact that as humans we do appreciate some feedback when it comes to handling objects. The answer to this conundrum is the concept of a tangible user interface.

In essence, a TUI allows users to interact with a digital system by manipulating physical objects linked to the particular system with a certain set of rules. For example, imagine a computer system in which the physical environment is seamlessly melded with the corresponding digital domain that recognizes the manipulation of the real world object in real time. The most primitive form of a TUI is the humble computer mouse. In fact, moving a mouse across a flat surface to manipulate an on-screen pointer is the most direct method of interacting with a digital system through the manipulation of a physical object.

In today's relatively more modern tech era, we have much more awe-inspiring examples of TUI. Take a look at Microsoft Pixelsense, a computing surface that allows multiple users to interact simultaneously with objects placed on a large table-sized interactive surface. The platform can be programmed to sense the type of objects placed on the surface and carry out a set of commands accordingly. For example, you could place your phone on